

FoodGenome AI

Calibrated food recognition with conformal guarantees
and grounded nutritional retrieval

*What five negative results taught us about
when added complexity does not pay*

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food-red-omega.vercel.app

The problem

Photograph a dish. The system returns:

- the category, from 101
- a confidence that *means* something
- every candidate it cannot rule out
- nutrition traced to a USDA record

And, when it should:

nothing at all

A nutrition figure shown without qualification will be acted on.

A classifier restricted to 101 categories will return one of them for a photograph of a car.

So **knowing when not to answer** is a requirement, not a nicety.

Headline

97.16%

Food-101 test top-1
on a sealed split

99.56%

conformal coverage
1.54 candidates

97.94%

on accepted images
after abstaining on
2.3%

98.6%

RAG answers correct
100% grounded

Validation is a fixed 4% stratified slice carved from *train*, seed 1337. The 25,250-image test split was untouched until final evaluation — and the mask was hashed and verified before the fine-tune on separate hardware was allowed to start.

Method

Vision

- Three backbones run once, embeddings cached (~20 h)
- Every later experiment then takes *seconds*
- Small MLP heads, probability average

Reliability

- Temperature fitted on held-out data
- Split-conformal prediction sets
- Abstention on two signals

Knowledge

- 101 classes $\times$ 32 USDA nutrients
- BM25 + bi-encoder, fused by RRF
- Cross-encoder rerank
- Query rewritten with the predicted dish

Grounding

- Every number verified against a source
- Matched *per unit*
- Failures withheld, not annotated

The caching decision made rigour affordable

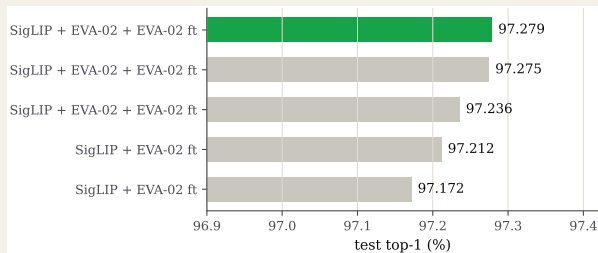
Running each backbone once and caching the embeddings cost twenty hours.

Every question afterwards became cheap:

- head architecture
- calibration, conformal sets, abstention thresholds
- **every ensemble combination** — rerun dozens of times

Without the cache each of those reruns would have been a fresh forward pass over 101,000 images. The ablation on the next slides simply would not have been affordable.

Negative result 1 — a trained head lost to arithmetic



A gated fusion head:

- 3.97M parameters
- 21.9 minutes of training
- **96.97%**

A 50/50 average:

- 0 parameters
- milliseconds
- **97.16%**

McNemar: the average wins at $p = 0.005$. The fusion head does *not* beat its own best input ($p = 0.071$).

Negative result 2 — the third backbone earned no place

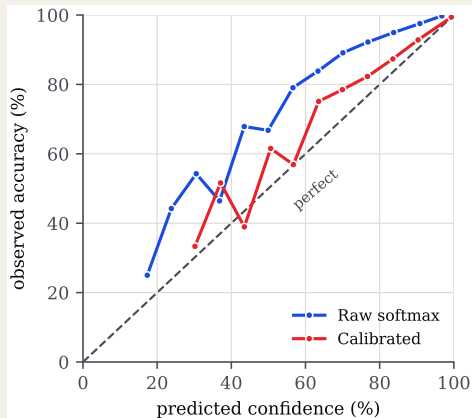
DINOv2-L is the only **self-supervised** member, so it was predicted to make the most independent errors.

It makes the least:

Share of SigLIP's errors also made by...	
EVA-02-L (supervised)	65.2%
DINOv2-L (self-supervised)	66.3%

Adding it moved the ensemble to 97.09% — marginally *below* the pair, at $p = 0.312$.
A weight search on validation independently drove its weight to **0.05**.

Confidence that means something



The ensemble sat **above** the diagonal — systematically *under*-confident, the less common direction.

	before	after
ECE	0.0514	0.0062
mean confidence	92.0%	96.6%
accuracy	97.16%	97.16%

Accuracy is unchanged and must be — temperature scaling is monotonic. It changes only what the number *means*.

A guarantee, not a feeling

A calibrated confidence is a claim about one image.

A **conformal set** is a claim about the procedure — and it holds without assuming anything about the model.

Target	Method	Coverage	Mean set size
90%	LAC	93.59%	0.95
90%	APS	90.18%	1.54
99%	LAC + top-1	99.56%	1.54

Below the model's own 97.16% accuracy the guarantee is met by the single best guess and the “set” degenerates to one class. The machinery only does work at 99%.

Negative result 3 — the broken version looked *better*

An early implementation forced the top-1 class into the set at test time, but scored **without** it during calibration.

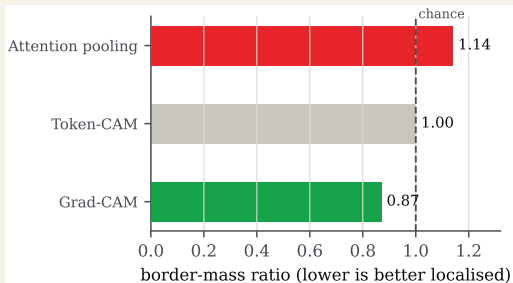
That silently rescued the 13% of images whose top score exceeded the threshold:

target **90%** → measured **99.1%**

The guarantee was **void** — and the reported number was better for it.

Conformal validity requires the identical rule in calibration and deployment. A bug that makes your metric worse gets found. One that makes it better does not.

Negative result 4 — attention is not saliency



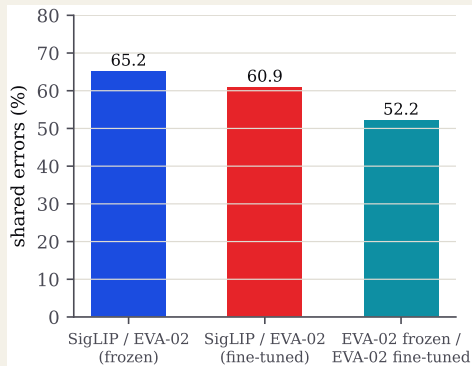
Share of attribution mass in the outer ring (60% of the frame), over 24 test images. Subjects are centred, so lower is better.

Reading the pooling head's own attention *should* be the most faithful account of where the model looked — those weights are literally the mixture being classified.

It is **worse than chance** at finding the food.

Reproduces the documented tendency of ViTs to park high attention on low-information background patches.

Negative result 5 — fine-tuning decorrelated but did not win



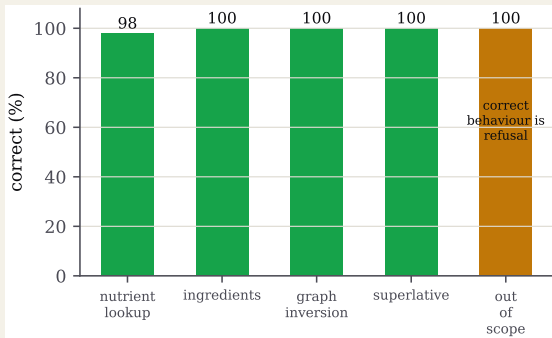
Nine GPU-hours: 6 epochs at 224 px, 1 at 448 px.

- Alone: 95.90% — *below* the pair
- At 224 px: 65.5% shared errors — no diversity at all
- At 448 px: **60.9%** — real diversity

The 448 epoch added **0.2** points of accuracy and **4.3** points of diversity.

Three-way ensemble: 97.27%, $p = 0.052$ — just short.

Retrieval that refuses



- 98.6% of in-scope questions correct
- **100%** refusal accuracy, zero false refusals
- 100% of answers grounded
- \$0.024 for the whole evaluation

Ground truth is *derived* from the knowledge base, not judged by a model — a language-model judge shares error modes with the system under test.

The grounding gate caught a real one

First live query. Asked for sodium per serving, the model answered:

983.4 mg

That figure is **arithmetically correct** — 447×2.2 — and the source even states the multiplier.

But it appeared in *no document*, so it was rejected.

The fix went in the corpus, not the tolerance. The micronutrient documents stated only per-100 g, so the model had to multiply to answer at all. Both bases are now written explicitly.

Loosening the check would have hidden exactly the class of error it exists to catch.

What it cannot do

- **Attribution is diffuse** — $0.87\times$ chance. The classifier sees a globally pooled vector, so position is discarded before it ever sees the image.
- **60 of 101 nutrition profiles are composed** from weighted ingredients, not measured. Every surface says which.
- **Abstention is a baseline**, not a trained OOD detector — it reports the model is lost, which is not the same as “this is not food”.
- **The knowledge base is closed** at 101 categories. Anything else is refused rather than estimated.

Oracle accuracy — the ceiling if a perfect combiner always picked a correct member — is 98.43%. Only 1.57% of test images defeat every model, and much of that is Food-101’s own label noise. This approach is close to exhausted.

What the negative results have in common

Four of the five share a shape: **an added component had nothing left to improve.**

- The fusion head had nothing to learn — two correlated members need no per-input weighting.
- DINOv2 added no diversity — self-supervision did not produce independent errors here.
- Fine-tuning added diversity, but too little to move a 25,250-image comparison.

Reporting them costs a headline number and buys a defensible account of where the remaining headroom *is not*.

Thank you

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github.com/A-K-M-Asifuzzaman/Food

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